

Commercial Systems Development in a Changed World

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The world has changed dramatically in the past ten years due to many political, technical, and cultural imperatives. One of the key areas both affected by these changes as well as the center of the transformations is the ability to transport people and cargo quickly, efficiently, and safely anywhere in the world. Modern jet transports represent one of the largest and most complex design problems known to modern society. The systems engineering discipline offers ways to manage size and complexity and assure resulting excellence in engineering.

This paper along with companion papers examine the subject of how the most recent of these airplanes, the Boeing 777, is developed and outlines the advances that were made to the systems development process. Systems engineering concepts get tested to their limits on commercial transports. Because business issues including cost and cycle time must be met on commercial projects, this paper proposes that commercial systems engineering, especially on large jet transports, provides valuable insights and lessons learned about systems engineering as an entire discipline for many types of commercial and defense applications.

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SYSTEMS ENGINEERING ON A GRAND SCALE

The 777 is the most recently introduced product in the field of large, commercial, jet transport airplanes. It consists of 66 separate yet integrated systems and has been the subject of the popular PBS television series and numerous technical publications and awards (e.g., the 1995 National Aeronautics Association Robert J. Collier Award). It represents systems engineering on a grand scale. In the five papers that constitute this series, this first paper provides a macroview of the context of modern systems engineering and the achievements of the 777. The four companion papers describe the systems engineering processes and lessons learned in the development of two of the most significant 777 systems, the AIMS (Airplane Information Management System) and the Flight Controls System (including the fly-by-wire functions), by those engineers and managers that led the teams. The suppliers for these systems were Honeywell Air Transport Division and Rockwell Collins (for part of the flight controls), respectively. The systems engineering papers are arranged to first provide the Boeing viewpoint and then the supplier's perspective. In total this set of papers provide a progressively deeper look into how modern, complex systems are developed along with the challenges and the successes.

WHAT IS SYSTEMS ENGINEERING?

Systems Engineering. What does this mean? There is much industry debate on this issue and more definitions than an airplane has rivets. Yet surely at its heart it is the application of good engineering principles to the development of a system within schedule, business, and market need constraints. It involves conceptualization, design, test, approval/certification and the fielding of a system. In this paper "systems engineering" is used in a very broad context to cover the actions and processes performed to engineer a product at the many levels required for a large development effort. A greater formalization of the discipline has become more urgently needed as the size and complexity of the systems increase and as the number and diversity of "stakeholders" increase. In essence, systems engineering is a problem-solving discipline for the modern world.

A CHANGED WORLD ORDER

For years the prime proponent of systems engineering was the Department of Defense for military systems. This was due to the need of the U.S. to apply the highest capability (technology, integration, and interoperability) to assure offensive and defensive advantage. Technical performance was

key. However, in the past two decades and especially in the last six years, several global factors have come together to shift the need for systems engineering into a much broader range of applications strongly focused in the commercial markets.

The first factor was the rapid, widespread growth of digital technologies resulting in a doubling of capability every eighteen months for the same capital investment (Moore's Law). This technology enabled affordable "intelligence," control, and integration to penetrate many commercial application areas, such as commercial aviation, automotive, medical devices, desktop computing, etc.

The second factor was the end of the cold war, resulting in realigned national priorities for the former adversaries, especially wherein defense dollar value and "business"-based performance became as important as technical performance.

The third factor was the rapid globalization of world markets and economies thereby totally altering international business relationships, competition, and security issues.

The end result of these changes was fourfold: 1) systems became more capable, complex, and integrated, 2) business value-based considerations drive everything from defense competition for its share of national budgets to company-based competition for market share and profits and country-based competition for standard of living and currency value, 3) the margin for error in decision making has greatly tightened, causing a greater need for optimization and risk reduction, and 4) the speed of everything is increasing.

All these factors have greatly increased the need for systems engineering, especially in the commercial markets, and have increased the motivation of the Department of Defense to adopt commercial practices and procedures for developing defense products along the lines of economies that drive commerce. For example, in order to obtain the efficiency benefits of commercial systems development practices and procedures, in 1994 Dr. Perry, then Secretary of Defense, issued a key memo eliminating the enforced reliance of military contractors on Mil Specs.

LARGE COMMERCIAL JET TRANSPORTS

One of the applications of systems engineering where all these world forces come together is large commercial jet transports. By any measure these products are one of the most complex, technology-based, engineered systems known to modern society. The development costs are so high, the skill levels of all types so wide, the international cooperation and coordination so broad and the risks so great that only three major companies currently compete against each other. They are The Boeing Company, Airbus (a European consortium of four

companies and countries), and McDonnell Douglas. (See note on p. 636.) As a whole, these products are awe inspiring to both the public as well as the people that design, build, and operate them. Consider that something the length of a football field costing more than 870 average homes (about seventy-five city blocks worth) and consisting of millions of parts could lift off the ground and routinely and safely fly hundreds of people per flight a total of millions of yearly miles anywhere in the world!

These airplanes exceed any measure of goodness by which one could judge the quality of a system's design and hence, the systems development process. Consider a few of these. Commercial transports are developed cooperatively between airline customers, regulatory agencies, manufacturers, and suppliers. Although the competition is intense among all the companies, they also work together to assure industry safety and all manner of standards to reduce the cost of the airplanes and support services and improve commonality. Once an airplane program is launched the products are delivered nearly half a decade later *on the day* committed and operating within the expectations of the buyer. They make a profit for all companies involved. They satisfy the end customers' needs (passengers and cargo companies) and are totally accepted and relied upon by these customers.

In spite of their complexity and the potential of high risk for flight, they are the safest form of transportation by at least two orders of magnitude. For example, in the United States there are approximately 41,000 fatalities per year due to motor vehicles, yet large commercial transport airplane accidents account for an average of only 72.4 fatalities per year (1990–1994). Translated into the key parameter of fatalities per 100 million passenger miles, the rates are 0.82 for motor vehicles versus 0.01 for all scheduled airlines (includes commuters which raise the rates). (All data quoted from the National Safety Council's "Accident Facts 1995".)

Now, what other products or systems can claim this? Think of delayed military, software, and civil projects, for example, many of which are outright canceled due to nonperformance for technical or business factors.

THE BOEING 777

The most recent transport airplane to be developed is the Boeing 777. Many advances to the practice of systems engineering and systems development were introduced on this airplane, including new tools, processes, teaming arrangements, safety improvements and technology all designed to give the airline buyers and end customers, the flying public, increased comfort, features and dollar value.

Perhaps the most spectacular change of all was the huge advances brought about by digital

technologies both in the design of the airplane and in the systems on the airplane itself. This airplane has over 100 Mbytes of very closely integrated and in many cases safety-critical software. The embedded software monitors and controls major functions throughout the airplane such as reconfigurable glass cockpit displays, fly-by-wire digital automatic landing systems, central maintenance functions, satellite communications and “PC-under-the-seat” passenger entertainment systems.

The systems interoperate with each via new technology digital data buses (ARINC 629 standard) designed for ultra safe two-way communication. The display system receives thousands of signal inputs from all the other systems on the airplane for display to the pilots. The central maintenance system receives tens of thousands of signals from all the systems to monitor and record maintenance related status and failure conditions that assist ground mechanics in quickly isolating faults and replacing equipment.

Often military systems are held in high esteem for advancing technology limits and complexity, as well they should. However, commercial systems have in many cases gone beyond military products in these advances. For example, there is more software currently flying on a commercial 777 than will be installed on the F-22 air superiority fighter for the twenty-first century, has not yet even completed development! It should be noted in this comparison that the F-22 has the largest software load of any fighter in history.

SYSTEMS PROCESS IMPROVEMENTS

In design methodology and processes Boeing and its suppliers took major steps forward in several areas. If “process” can be thought of broadly as how tasks are done, then it includes teaming arrangements of who is involved in the design as well as how and when, the methods used, the deliverables and data required, the tools, and the facilities. Standards and guidelines are often needed to assure compliance, organization and commonality for projects this size.

On the 777, systems development advances included new “Working Together” teaming relationships with airline customers and suppliers; ETOPS (extended time operations) at initial certification; a state-of-the-art CAD/CAM system called CATIA; improved, consistent systems requirements and design capture approaches; a massive, new integrated test facility; and what is called a “Service Ready” airplane. All of these advances had never been accomplished before on a single product anywhere near this size and scope in the industry. These are outlined below.

Working Together: Working Together was a Boeing and airline term for concurrent engineering.

The program brought together teams consisting of engineering, key airlines, suppliers, manufacturing, and customer service organizations all at the front end of the program to define the requirements and design. This was a successful way to address the classical systems engineering failures of defining an incomplete design, one unbuildable or one that doesn’t meet the customers requirements. “Working Together” teaming has now become the standard way of doing business for Boeing.

ETOPS: ETOPS is a commercial aviation term for twin engine airplane operations at long distances from a diversion airport. The concern is the failure of the second engine when there is only one left. Normally this special type of operation is granted by the governmental certification agencies only after two or more years of in-flight operational experience to prove the airplane safety and address design changes where needed. On the 777, a new systems development approach was introduced wherein new requirements management, ultrahigh design quality and an extra year of flight test under actual operating conditions were added to the program.

CATIA: Airplane designs are implemented in hard parts (structure, tubing, ducting, wiring), electronics, software, electrical, and mechanical units. For hard parts, the airplane’s geometry design, one of the most major process changes in Boeing’s history was made. The CATIA CAD/CAM system (a product developed by Dassault of France and IBM) was purchased, extended for large scale use and integrated into Boeing’s engineering community, its factory, its customer service organization, and its suppliers.

This was an enabling technology for Boeing requiring many organizational changes, process changes, a huge computing support structure and a fundamental rethinking of jobs and relationships in how the company develops, builds, and supports airplanes. Historically, an airplane design is normally translated into full scale mockups (i.e., a full size airplane) to assure engineering integration of all the separate systems, to determine errors in the engineering drawings and to correct them. This is now all done in the CATIA computer system and network through digital preassembly (DPA) which allows the engineers to assess design, fit, and interference directly from the engineering CATIA datasets. This eliminated the need for mockups. Furthermore, all the more than 10,000 datasets that define the airplane are available on-line in the latest release to everyone. This enables the customer services organizations that provide training and manuals to directly download the three-dimensional geometry datasets into two-dimensional vector graphics files and use these figures directly in flight maintenance, and support manuals as well as computer-based training. This “macroprocess” change has greatly reduced

rework across the organization, decreased cycle time for changes, and significantly cut errors for geometry information between engineering, the factory, and customer services.

Most of an airplane's geometry design is in structures. However, of total airplane development costs structures represents only 28%. The largest portion of nonrecurring costs is for airplane systems (avionics, flight controls, environmental control, mechanical, hydraulics, electrical). These cost 54% of the total. (The other costs are 5% for aerodynamics, 7% each for propulsion and payloads, and 6% for the remainder.)

System Requirements and Design Capture: To address the airplane systems, additional changes were required beyond CATIA. Prior to capturing design implementation details (such as the CATIA geometry definition) the system itself must be defined. Especially significant are the definition of system operations, functions, and interfaces. To define these nongeometry aspects of the 777 systems, new processes were added to the program for capturing operations, functions, and interfaces in three key areas: system requirements, system architecture and designs, and system safety analyses. These new processes were governed by guidelines developed by cross-functional teams.

On previous programs, systems and their system development processes were less integrated than on the 777. Each system group defined its own systems engineering approach. Although this produced excellent airplanes, it was inefficient, thereby raising development costs, errors, and rework. It also made it difficult for downstream users of engineering design data to understand the system's design, operations, or functions.

On the 777, consistency was attained by assuring that all groups defined requirements and designs in consistent deliverables. Requirements were captured in Systems Requirements and Objectives deliverables; and architectures and designs were captured in Systems Design Description deliverables. In each of these, the deliverables are structured around an information model that provides traceability across all levels of system requirements and all levels of system designs as well as traceability between requirements and designs. This step on 777 is paving the way to the future and next program to use more model-based systems engineering and consistent, single-source, sole-authority digital datasets for the system operations, functions, interfaces, and implementations much as was done for the geometry design with CATIA datasets.

IASL: In order to assure verification of the performance and integration of the 66 subsystems on the 777, a new test facility was built called the Integrated Airplane Systems Laboratory (IASL) in Seattle. This is the largest facility of its type in

the world. Although initially focused on the 777, it is designed to test and integrate avionics, flight controls, electrical, mechanical, and crew systems for all systems for Boeing's entire fleet of airplanes. In sheer physical terms, it has 4.7 acres of roofing, 520 m of electrical wire, 160 m of data cabling, 270 tons of doors, and 518,000 ft² of total space.

The IASL has three main functions: 1) pilot interface development, 2) standalone testing of individual subsystems, and 3) integration testing of all airplane systems. During the design phase of a program its "Cabs" (the flight deck of an airplane) and simulators are used to define and validate the requirements for the flight crew displays and controls. For standalone testing, the engineers test their individual subsystems with their interfaces to other units simulated. For the final program phases of a program the IASL can test all systems interconnected as they would be on a real airplane with the pilot in or out of the loop. The IASL was used to "fly" test flights of the 777 before it even rolled out of the factory! It was an important element of the overall process to assure early ETOPS certification for the 777.

Service Ready: Our airline customers require more than just excellent airplanes. Their operations require documentation, training, and support to assure efficient and safe use of the airplanes. As airplanes enter service, reliability increases as all these support services and operations mature. On the 777 it was desired by the airlines to speed this process. In order to receive early ETOPS certification, it was also required by the FAA and its European equivalent, the JAA, to have airplane support mature at the time the 777 entered service. We called this "Service Ready." This was an entirely new process to have customer service deliverables a part of the engineering certification. Part of the engineering steps that assisted this were the CATIA datasets and the Systems Design Description datasets both of which gave the customer services organization better, earlier, and more consistent information than they ever had before. The 777 in-service dispatch reliability is currently 97.7% and climbing. This is the highest of any Boeing airplane at this early date in its operational life thereby indicating the success of the service-ready concept. Service ready is now becoming a standard practice on all Boeing programs.

All these changes mentioned above are moving gradually toward corporate, shared, sole-authority, digital datasets of product design and support information. Both for the geometry information and for the systems nongeometry information the systems engineering process within the Boeing Commercial Airplane Group has been structured around this shared data concept rather than around traditional "waterfall", task-based lifecycle systems engineering models.

CONCLUSIONS AND FUTURE DIRECTIONS

Airframe manufacturers such as Boeing along with system suppliers, the airline industry, and the regulators have continued to introduce improvements to the efficiency, service, and safety of air travel through the use of teaming, technology, integration, process changes and engineering excellence. It is the entire enterprise that has fine tuned these achievements.

The public has been served well and more is coming. The electronic revolution has not yet even hit middle age and will provide incredible new capabilities that will increase several types of efficiencies and passenger features. Examples include: 1) new passenger entertainment, communication, and business services some of which are now being introduced on the 777, and 2) large economic improvements to airline operation and the efficient use of the entire airspace via communication, navigation, and surveillance (CNS) capabilities implemented through improvements in Air Traffic Management (ATM), satellites and ground systems.

Business values such as reduced airplane development costs and reduced design cycle time will need to be addressed in much deeper ways as we all continue to adjust to deregulation. All of this will require further improvements to our

systems engineering processes, especially an increased integration and sharing of the total range of data as the center piece for this process. Additionally, deeper cost and business value considerations need to be incorporated into the daily engineering trade studies and decision making to meet market requirements.

In no small part these successes are due to the engineering processes, standards, and experience that have developed in this industry. Although it often goes by different names in the commercial world than the defense world, the systems development process *is* the systems engineering process. These processes have responded to modern business and international pressures over years and could have wide applicability to assist the engineering effort in many other fields, such as automotive, medical, defense and many others. Similarly, there is much experience in these fields that is now critical to commercial aerospace such as global communications and PC-based LAN networks. Cross-industry fertilization of experiences would be very healthy for all concerned and would lead to a deeper development of the systems engineering discipline and the needed twenty-first century tools for systems development.

Note added in proof: As this paper is being written, it is likely that Boeing and McDonnell Douglas will merge, thereby leaving only two competitors.



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He is the past president of the Systems Development Forum (SDF), a Governor of the IEEE's Aerospace Electronic Systems Society, is general chairman of the upcoming 17th Digital Avionics Systems Conference (DASC) and is a charter member and local director of the International Council on Systems Engineering.